Linear Regression [1]

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[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

Linear Regression – a fundamental learning algorithm

- Supervised learning method
- It assumes the dependence of Y on X is linear
- Largely used in many disciplines
- Simple and interpretable
- Fundamental in data science

Linear Regression Models

- Simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Multiple linear regression

$$Y = \beta_0 + \beta_1 X + \ldots + \beta_p X_p + \epsilon$$

- Y: dependent variable (response, outcome)
- X's: independent variable (covariates, explanatory variable)
- β's: regression coefficients
- ϵ: random error (irreducible error)

Linear Regression Models

- Using matrix format

$$Y = \mathbf{X}\boldsymbol{\beta} + \epsilon$$

- $\mathbf{X}$ is called design matrix with first column being 1's
- The *estimated linear regression model* is

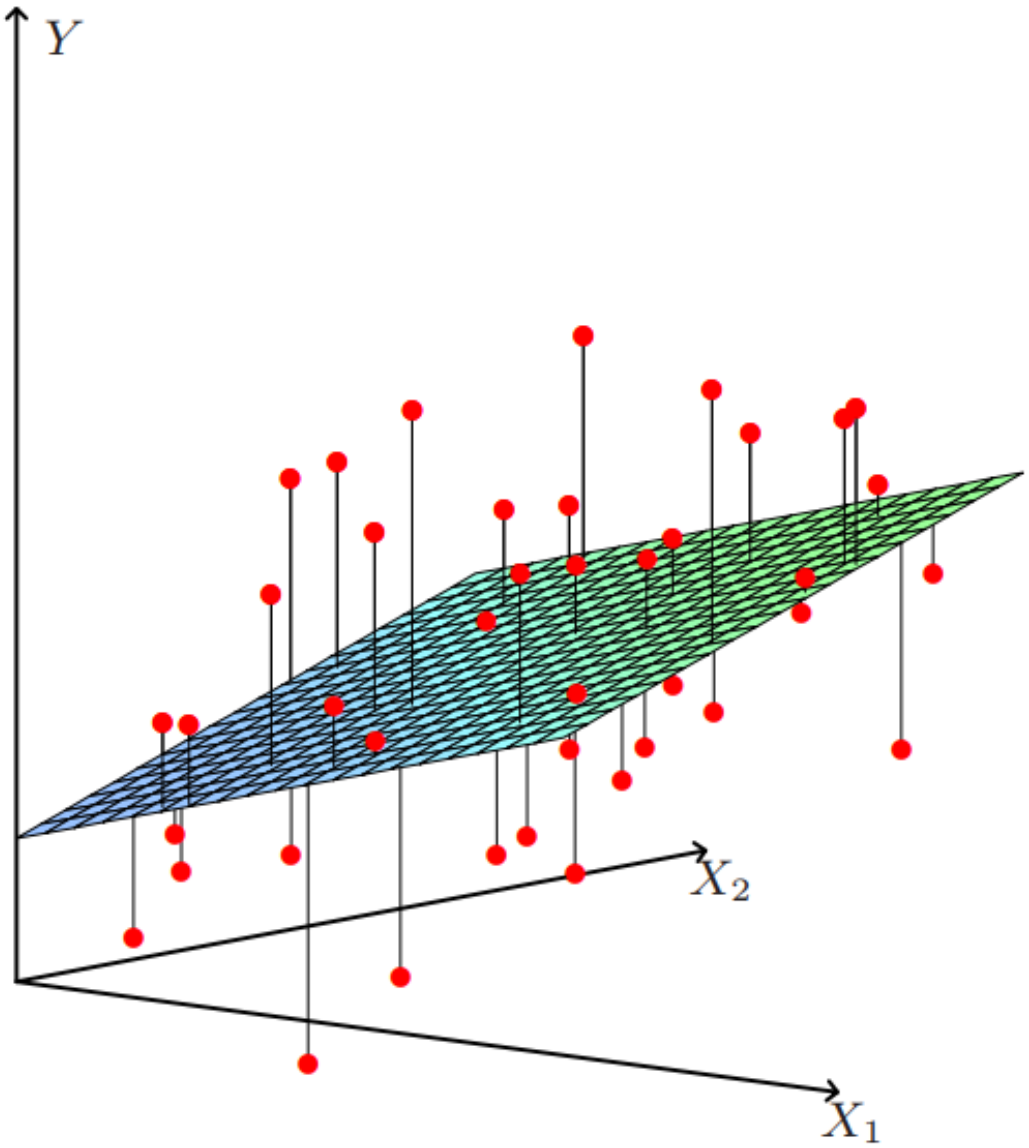
$$\hat{Y} = \mathbb{E}(Y|X) = \mathbf{X}\hat{\boldsymbol{\beta}}$$

- Goal: estimate regression coefficient β

Model Assumptions

- $\mathbb{E}(Y|X)$ is a linear function of X or its basis expansion such as X_1^2, $X_2^3, \ldots$
- The error term $\{\epsilon_i, \ldots, \epsilon_n\} \overset{i.i.d.}{\sim} N(0, \sigma^2)$

Least Square Solution



Least Square Solution

- We want to minimize residual sum squares

$$
\begin{aligned}
RSS(\beta) &= \sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \beta)^2 \\
&= (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)
\end{aligned}
$$

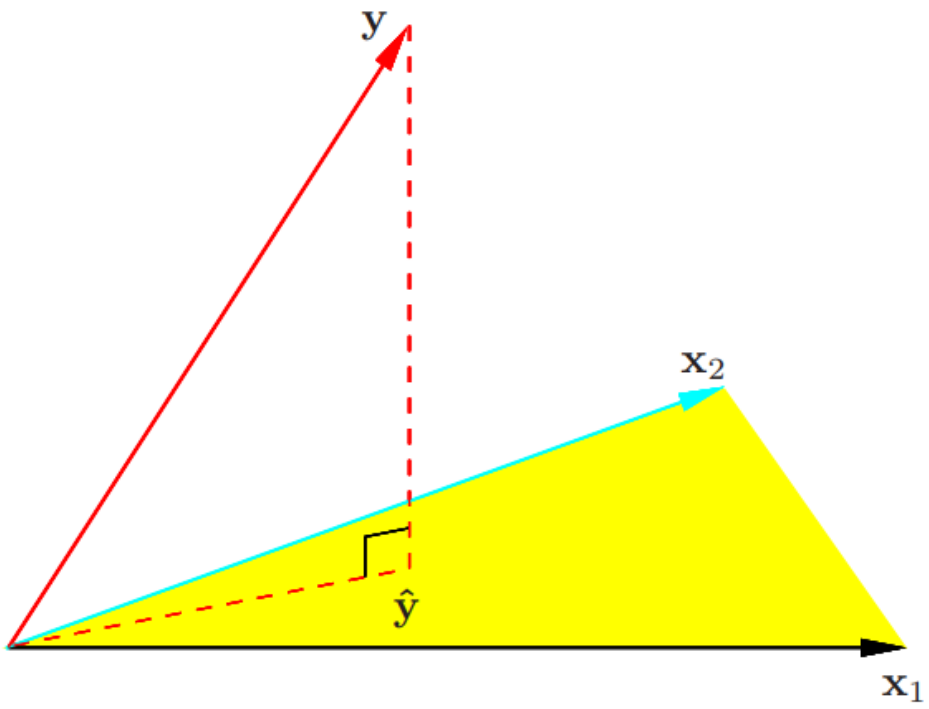
- Take first-order derivative with respect to β and set to 0

$$
\begin{aligned}
0 &= \frac{\partial RSS(\beta)}{\partial \beta} = -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) \\
\mathbf{X}^T\mathbf{y} &= \mathbf{X}^T\mathbf{X}\beta
\end{aligned}
$$

- This is called *normal equation*.

Least Square Solution

- By assuming $p < n$, the solution is $\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
- The predicted value is $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
- $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is called hat matrix or projection matrix



Important questions

- Is at least one of the predictors useful to predict and explain the response?
- Adjusted Do all predictors help to explain response, or just a subset?
- How well does the model fit the data?
- Given a set of predictor values, what response value does the model predict? How accurate is the prediction?

Model Assessment – R Square and MSE

- It is proportion of variation in Y explained by the model

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

  R^2 increases monotonically as number of explanatory variable increasing.

- Adjusted R^2

$$R^2_{adj} = 1 - \frac{n-1}{n-p-1}\frac{RSS}{TSS}$$

- Mean Squared Error (MSE)

$$MSE = \frac{1}{n-p-1} \times RSS$$

  It is an unbiased estimate of σ^2 for irreducible error ϵ.

Model Assessment – AIC, BIC, Cp, CV

- Akaike information criterion (AIC), the smaller the better

$$AIC = -2\log(\hat{L}) + 2p$$

- Bayesian information criteria (BIC), the smaller the better

$$BIC = -2\log(\hat{L}) + \log(n)p$$

  where $\hat{L}$ is estimated likelihood function
- Mellow's C_p is the same as AIC for linear regression
- Cross-validation error (CV score)

Hypothesis Testing – Test individual coefficients

Is a specific X relevant?

- Testing for individual β
  - H_0: $\beta_j = 0$; H_1: $\beta_j \neq 0$
  - Using T-test since the true variance is unknown

$$T = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p-1}$$

  where v_j is the jth diagonal element of $(\mathbf{X}^T\mathbf{X})^{-1}$
  - Reject H_0 if p-value $< \alpha$ or $|T| > T_{1-\alpha}^{(n-p-1)}$
- Confidence interval: $\hat{\beta} \pm se(\hat{\beta}) \times T_{1-\alpha}^{(n-p-1)}$

Hypothesis Testing – Test multiple coefficients

- F-test for overall significance
  - H_0: $\beta_1 = \ldots = \beta_p = 0$; H_1: at least one $\beta \neq 0$
  - F statistics

$$F^* = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$

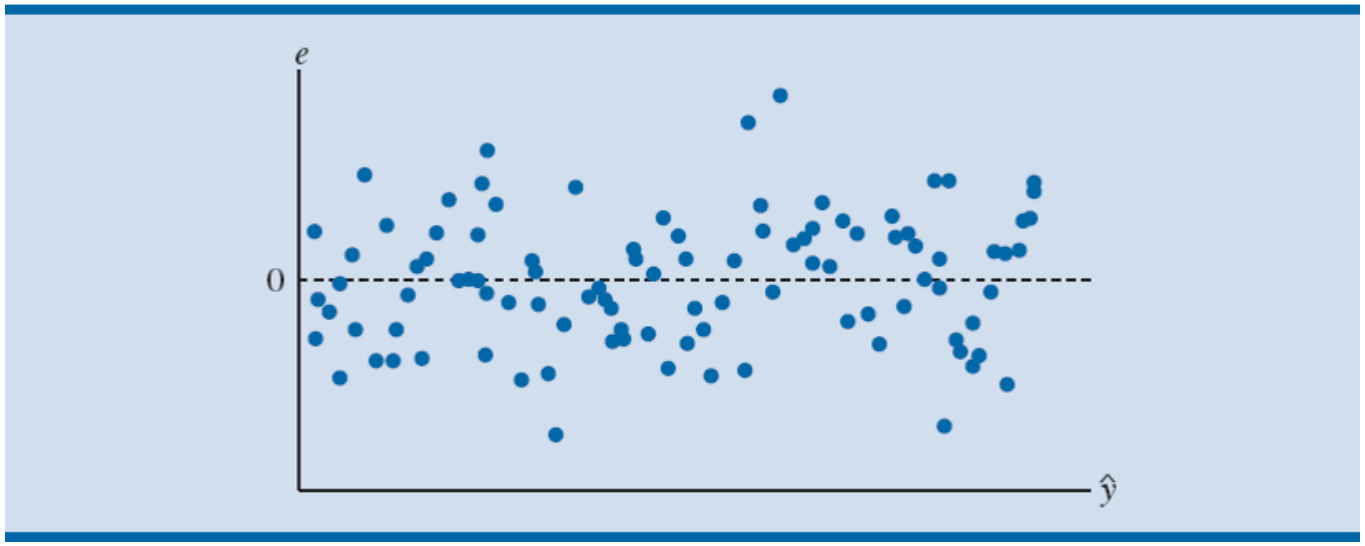
where $TSS = \sum_{i=1}^{n}(y_i - \bar{y})^2$, is total sum squares

Bootstrap

- Resampling method
- A powerful tool to quantify uncertainty
  - standard error
  - confidence interval
- Random sampling with replacement
- The general procedure:
  - fit a model **B** times based on **B** bootstrap samples
  - store all the parameter estimates
  - calculate standard error and confidence interval

Model Diagnostics

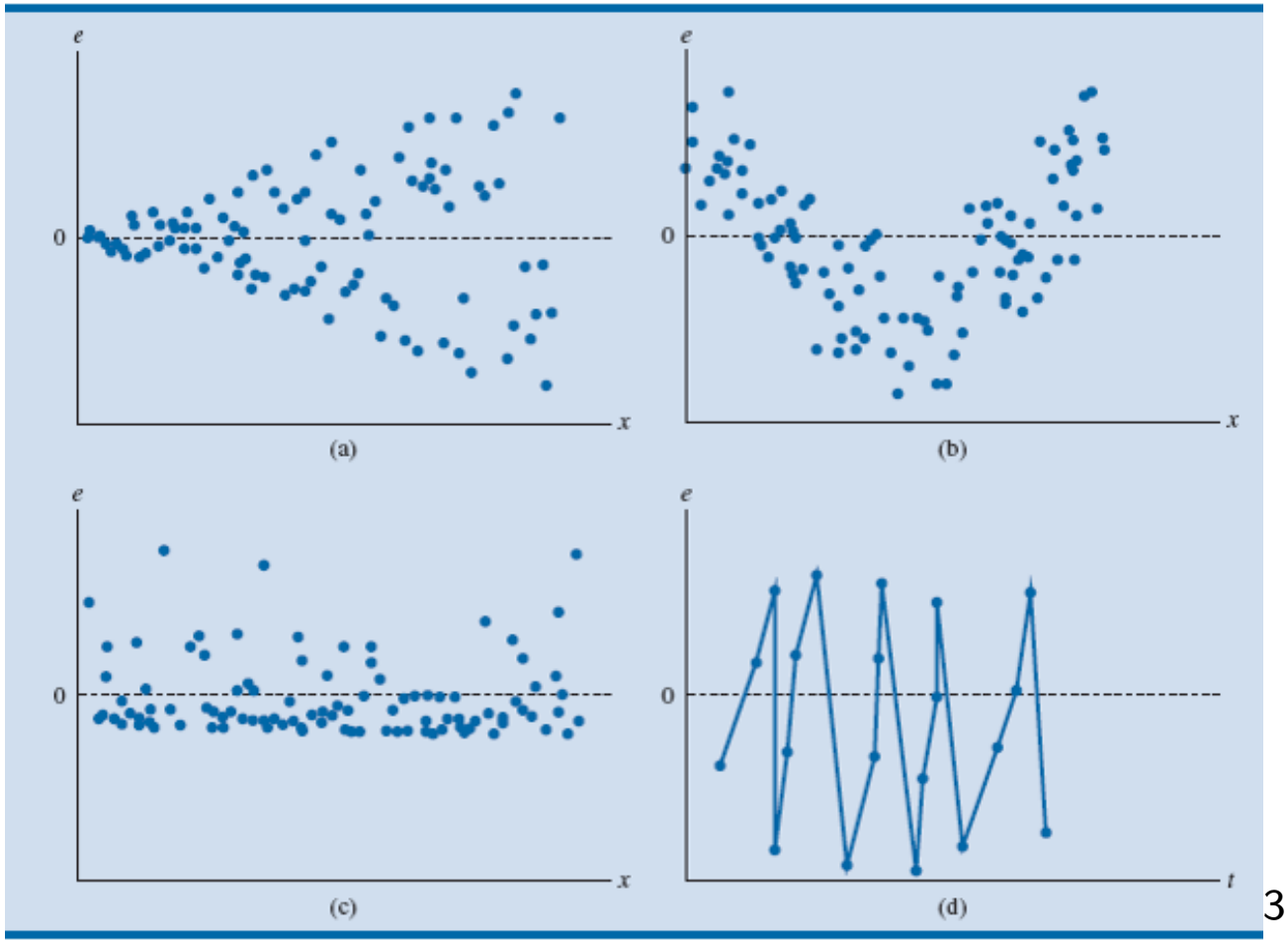
- Check the assumptions on the error term
  - Independent normal distribution?
  - $\mathbb{E}(\epsilon_i) = 0$?
  - $Var(\epsilon_i) = \sigma^2$ =constant?
- Residual plot (an ideal residual plot looks like this)[2]



[2]source: Camm, et al., *Essentials of Business Analytics*

Residual plot

Which type of assumption is violated?



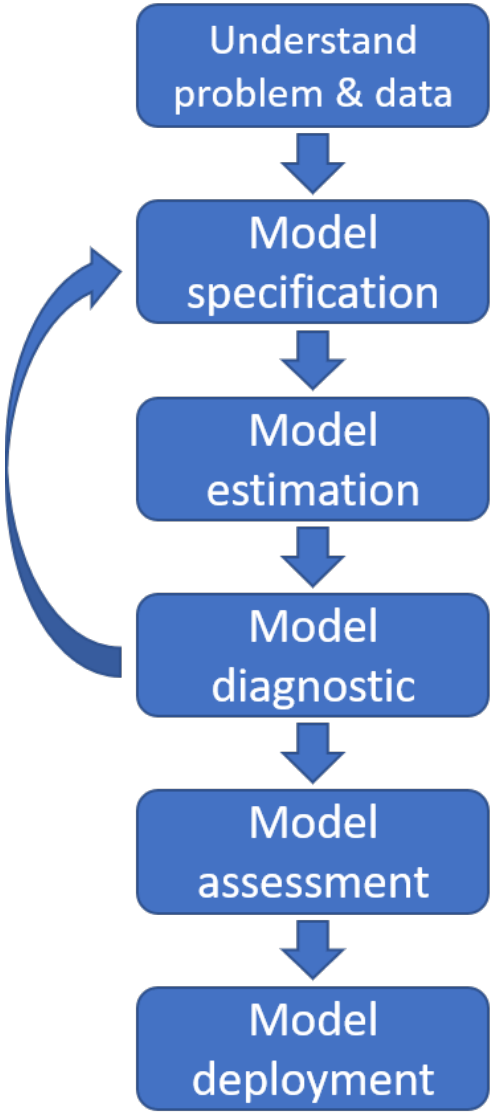
[3]

[3] source: Camm, et al., *Essentials of Business Analytics*

Other Diagnostic Plots

- Normal Quantile-Quantile Plot
  - It plots the standardized residual vs. theoretical quantiles
  - An easy way to visually test the normality assumption
  - If residual follows normal distribution, you should expect all dots lie on the diagonal straight line.
- Residual-Leverage Plot
  - This plot checks if there are any influential points, which could alter your analysis by excluding them
  - The points that lie outside the dashed line, Cook's distance, are considered as influential points

Model Building Process



- Describe the problem by stating the goal and purpose of statistical analysis.
- Given the dataset, conduct exploratory data analysis to maximize the knowledge of information carried by the data.

- Identify relevant variables and do necessary data cleaning and transformation. These include missing data imputation, variable recoding and transformation, etc.
- Identify appropriate statistical models to achieve the purpose of analysis.

- Use software to estimate the model. This process may involve debugging.
- Understand the main components of the output.
- Statistical inference needs to be conducted.

- Check if there is any model assumption violation. (For more ML oriented methods such as neural networks, this step is often skipped due to less assumptions.)
- If violation is identified, go back and re-specify the model. For example, if residual plots show quadratic pattern, then the quadratic term of one or more predictors needs to be included. Graphical inspections are needed in this procedure.

- Typically, various models are estimated, including a certain type of model with different sets of variables (determined by variable selection), and different types of model such as linear regression, regression tree, and neural networks.
- Compare all estimated models in terms of certain criteria, such as cross-validation error, AIC and BIC, etc.

- At the end of analysis, a final model needs to be determined for deployment.
- Depending on different scenarios, the final model may be a set of models that include several different models.
- Inference and model interpretation should be part of the deployment.